Buskirk, DASC 605

Student Computation Lab Number 1: Exploring Sampling Distributions

In your groups work through the various parts of this Lab. The results of the lab won't be submitted for credit, but be prepared to discuss your observations and answers to the Reflection questions.

Overview: Sampling Distributions of Common Statistics

This lab explores sampling distributions by repeatedly sampling from a right-skewed population of systolic blood pressure (SBP) values. You will study the behavior of the sample mean, sample median, and sample proportion, and examine how sample size and measurement quality affect variability and distributional shape.

Learning Objectives

- Explore the concept of a sampling distribution empirically
- Compare sampling distributions of the mean, median, and proportion
- Examine the effect of sample size on variability and shape
- Understand how measurement error inflates uncertainty
- Relate empirical results to the Central Limit Theorem

Context and Data

You are given a large dataset representing a population of adults undergoing routine health screening that was obtained from all members of the ForTheHealthOfIt health care plan in North Carolina. The variable of interest is systolic blood pressure (SBP), which is right-skewed for adults in this health system. While the dataset is about 80K cases, we are considering it as a large approximation to the true underlying population distribution that is theoretically unknown.

Population characteristics (known here for instructional purposes):
- Median SBP ≈ 125 mmHg
- Mean SBP ≈ 130 mmHg
- Approximately 33% of individuals have SBP ≥ 140 mmHg

Part A: Exploring the Population

1. Load the population dataset.
2. Plot the distribution of SBP.
3. Compute the population mean, median, and proportion with SBP ≥ 140.

Example R Code

```
sbp_pop <- read.csv("lab_sampling_population_sbp_diff5.csv")

mean(sbp_pop$sbp_mmHg)
median(sbp_pop$sbp_mmHg)
mean(sbp_pop$sbp_mmHg >= 140)

hist(sbp_pop$sbp_mmHg, breaks = 50,
   main = "Population Distribution of SBP",
   xlab = "SBP (mmHg)")
```

Part B: One Sample

1. Draw a random sample of size n = 30 from the population (with replacement).
2. Compute the sample mean, median, and proportion.
3. Compare these to the population values.

Example R Code

```
set.seed(1)
sample30 <- sbp_pop[sample(nrow(sbp_pop), 30, replace = TRUE), ]

mean(sample30$sbp_mmHg)
median(sample30$sbp_mmHg)
mean(sample30$sbp_mmHg >= 140)
```

Part C: Sampling Distributions

1. Repeatedly draw samples of size n = 30 (e.g., 1000 samples).
2. For each sample, compute the mean, median, and proportion.
3. Plot the sampling distribution of each statistic.

Example R Code

```
B <- 1000
means <- medians <- props <- numeric(B)

for (b in 1:B) {
  samp <- sbp_pop[sample(nrow(sbp_pop), 30, replace = TRUE), ]
  means[b] <- mean(samp$sbp_mmHg)
  medians[b] <- median(samp$sbp_mmHg)
  props[b] <- mean(samp$sbp_mmHg >= 140)
}

par(mfrow = c(1,3))
hist(means, main = "Sampling Distribution: Mean")
hist(medians, main = "Sampling Distribution: Median")
hist(props, main = "Sampling Distribution: Proportion")
```

Part D: Effect of Sample Size

Repeat Part C for n = 10, 50, and 100. Compare how the sampling distributions change in terms of spread and shape for each statistic.

Reflection Questions for Parts A - D

- Which statistic has the smallest sampling variability?
- Which sampling distribution appears most normal?
- How does right skewness in the population affect the mean versus the median?
- How does increasing sample size change the sampling distributions you observed?

Part E: Measurement Error Extension

In practice, SBP is not measured perfectly. Manual cuffs, rounding, posture, and clinician variability introduce measurement error. In this extension, you will examine how measurement error affects sampling distributions and how much larger a sample is required to achieve comparable precision.

We model measurement error as:

Observed SBP = True SBP + error, where error ~ Normal(0, σ_m^2).

Measurement Error Tasks

1. Set the measurement error standard deviation to σ_m = 8 mmHg.
2. For each sample, create a noisy version of SBP by adding random noise.
3. Recompute the mean, median, and proportion SBP ≥ 140 using the noisy measurements.
4. Construct sampling distributions with and without measurement error.
5. Compare their spread and shape.

Example R Code (Measurement Error)

```
sigma_m <- 8
B <- 1000
mean_true <- mean_obs <- prop_true <- prop_obs <- numeric(B)

for (b in 1:B) {
  samp <- sbp_pop[sample(nrow(sbp_pop), 30, replace = TRUE), ]
  x_true <- samp$sbp_mmHg
  x_obs <- x_true + rnorm(length(x_true), 0, sigma_m)

  mean_true[b] <- mean(x_true)
  mean_obs[b]  <- mean(x_obs)
  prop_true[b] <- mean(x_true >= 140)
  prop_obs[b]  <- mean(x_obs >= 140)
}

sd(mean_true)
sd(mean_obs)

par(mfrow = c(2,2))
hist(mean_true, main = "Mean (True)")
hist(mean_obs, main = "Mean (Measured)")
hist(prop_true, main = "Proportion ≥ 140 (True)")
hist(prop_obs, main = "Proportion ≥ 140 (Measured)")
```

Reflection Questions for Part E

- How does measurement error affect the center and spread of the sampling distribution of the mean?
- How does measurement error affect the sampling distribution of the proportion? Why is this effect pronounced near the threshold 140 mmHg?
- Which is more effective for reducing uncertainty: increasing sample size or improving measurement quality?
- Why can measurement error not be fully eliminated by increasing n?